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A Privacy-Aware Web Dashboard for Comparative Community Energy Analytics Using Simulated Smart Circuit Controller Data

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The problem: readings are not insights

- ✓Voltage, current and power readings do not explain consumption behaviour.
- ✓Households need load rankings, peak periods, cost estimates and fair comparisons.
- ✓Commercial controller APIs are not always accessible for research.
- ✓The research develops a testable interpretation layer, not a billing system.

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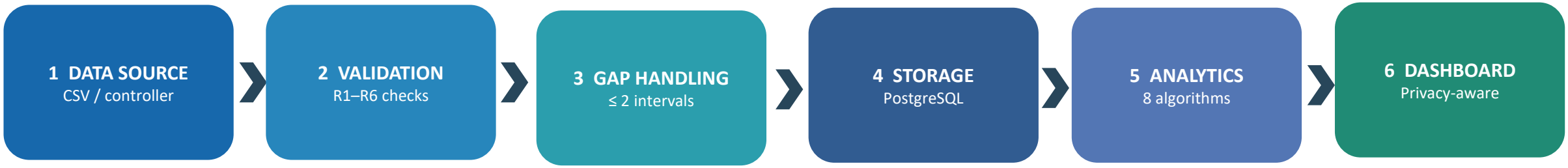
Research Aim and Objectives

- ✓Specify the electrical variables and dataset structure, including units and data-quality thresholds.
- ✓Design and implement a secure pipeline that ingests, validates, stores and pre-processes smart-circuit-controller readings.
- ✓Develop eight comparative analytics algorithms for baselines, peer comparison, peak detection, ranking, anomaly warnings and estimated cost.
- ✓Build a responsive web dashboard that presents individual and community-level insights without identifying other households.
- ✓Evaluate accuracy, completeness, responsiveness, usability and real-data compatibility.

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System architecture: six layers

- Simulated CSV now; approved API or local gateway later.
- Each stage has an independent quality or accuracy check.



Quality flags retained • Short gaps imputed • Audit log at each stage
Data source can be replaced without rewriting downstream layers.

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Experimental dataset and injected faults

- 10 anonymised sites × 6 channels × 30 days × 96 intervals/day
- 172,800 expected readings at 15-minute resolution
- 200 duplicates; 30 invalid voltages; 10 negative-power values
- 550 random losses plus a 96-reading communication gap
- 30 injected abnormal-use events per dataset; hold-out dataset B
- Note: reported raw ingestion count (172,343) requires CSV reconciliation.

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Validation and analytics methodology

- ✓Six checks cover schema, timestamps, duplicates, voltage, power and physical consistency.
- ✓Only short gaps (up to two intervals) are interpolated; longer gaps remain missing.
- ✓Eight algorithms calculate energy, cost, peaks, ranking, baseline, comparisons, completeness and warnings.
- ✓Community comparison is leave-one-out, not yet profile-matched.

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Data quality and numerical accuracy

240 / 240

✓Injected invalid records detected

1.63% -> 1.33%

✓Mean monthly error after interpolation

19

✓Unit tests reported as passed in Results

Independent recomputation agreed to floating-point precision; unresolved long gaps remain a source of error.

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Abnormal-use warning trade-off

Hold-out dataset B

Stage 1

✓Precision 0.49

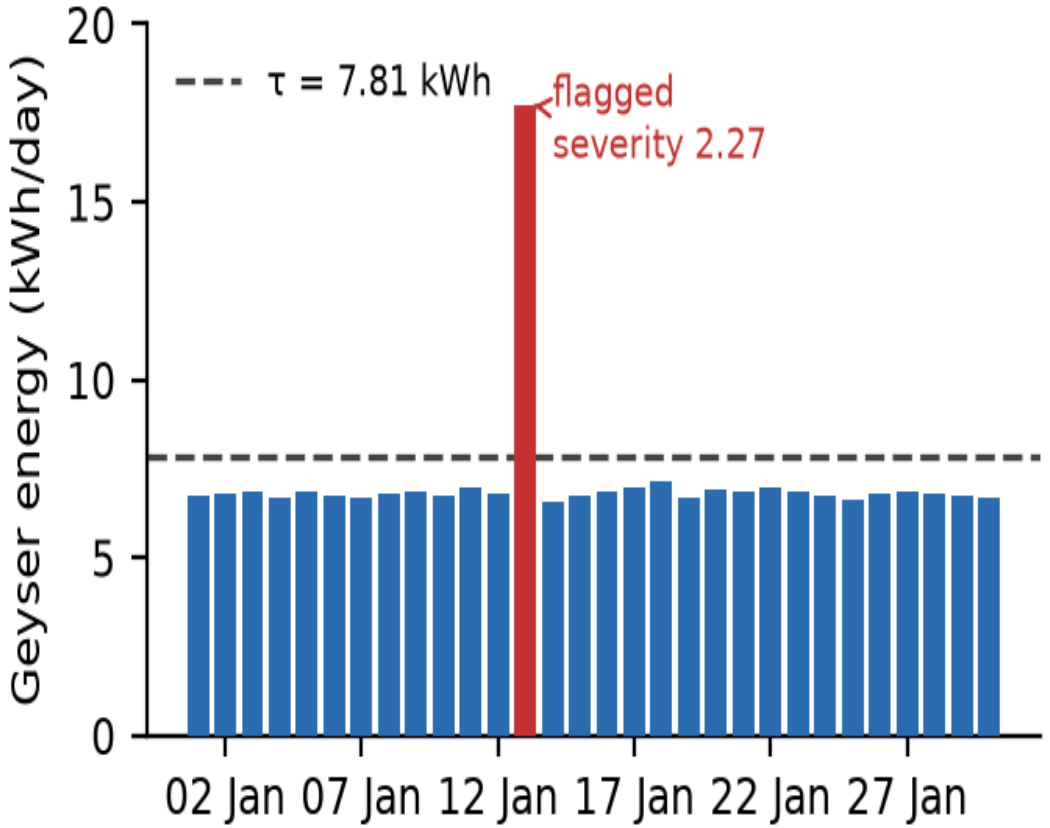
✓Recall 0.63

Stage 2

✓Precision 1.00

✓Recall 0.40

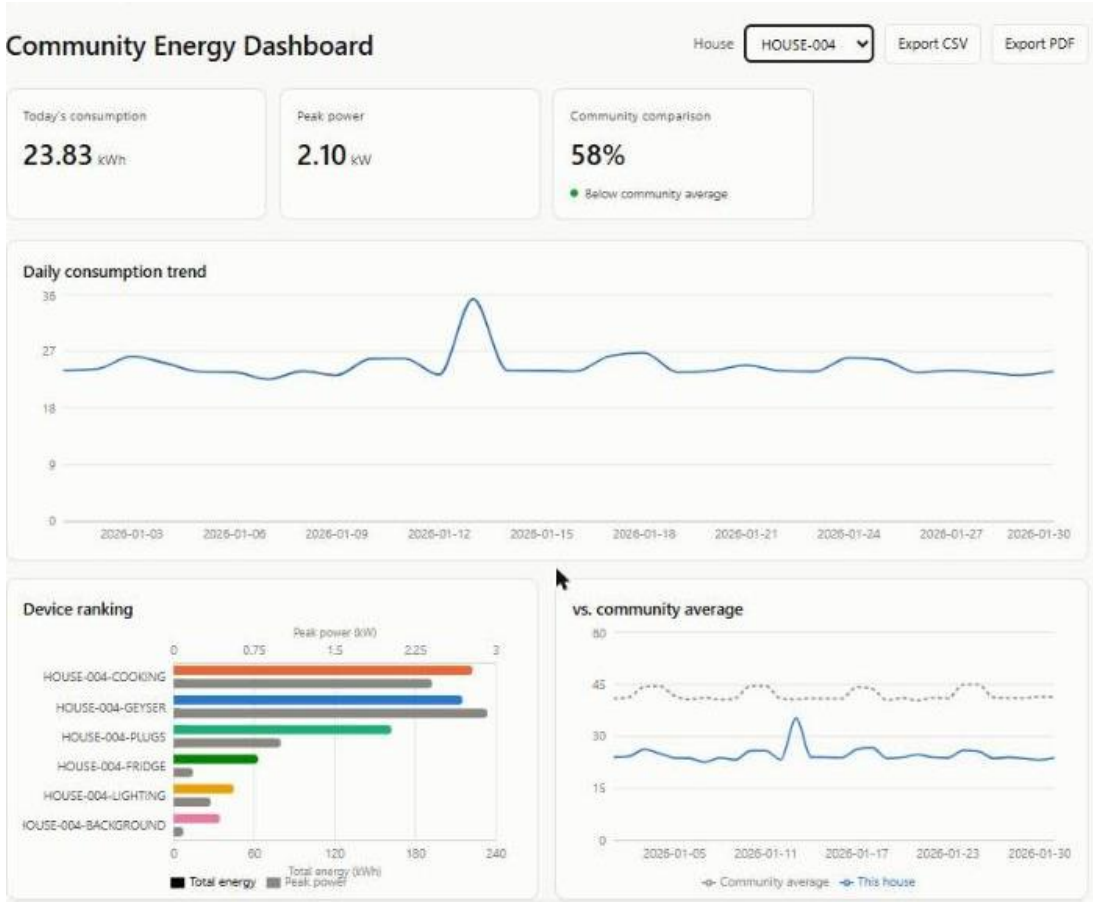
Stage 2 removed false positives but missed 18 of 30 injected events. Higher precision came at lower sensitivity.



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Dashboard and consumption insights

- ✓Site HOUSE-004: cooking 31.4%; geyser 27.9%; plugs 21.4%.
- ✓Peak use occurs around 18:00 in the simulated profile.
- ✓Resident view uses neutral status bands, not leaderboards.

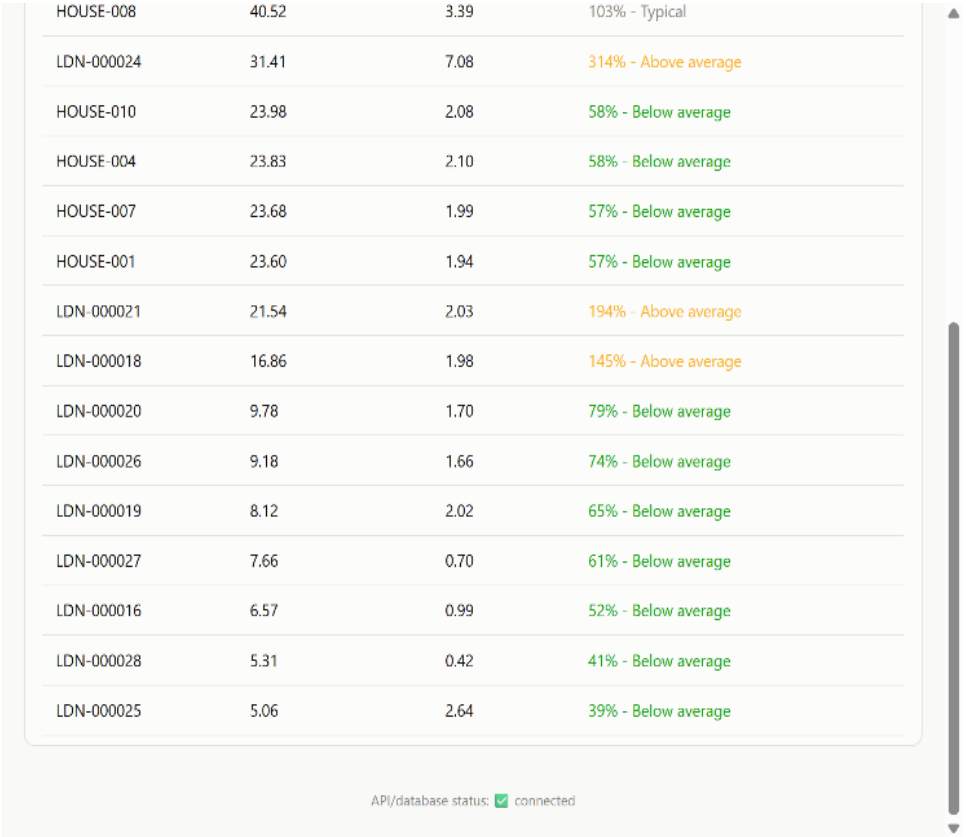


Source: live prototype dashboard (conference paper, Fig. 6)

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Validation with real smart-meter data

- ✓10 Low Carbon London households; half-hourly whole-house readings.
- ✓14,400 valid records and 10 duplicate records identified.
- ✓Monthly household totals ranged from 134.5 to 839.0 kWh.
- ✓Community comparison ratios widened to 0.39-3.14.
- ✓This confirms source compatibility, not real CBi device integration.



HOUSE-008	40.52	3.39	103% - Typical
LDN-000024	31.41	7.08	314% - Above average
HOUSE-010	23.98	2.08	58% - Below average
HOUSE-004	23.83	2.10	58% - Below average
HOUSE-007	23.68	1.99	57% - Below average
HOUSE-001	23.60	1.94	57% - Below average
LDN-000021	21.54	2.03	194% - Above average
LDN-000018	16.86	1.98	145% - Above average
LDN-000020	9.78	1.70	79% - Below average
LDN-000026	9.18	1.66	74% - Below average
LDN-000019	8.12	2.02	65% - Below average
LDN-000027	7.66	0.70	61% - Below average
LDN-000016	6.57	0.99	52% - Below average
LDN-000028	5.31	0.42	41% - Below average
LDN-000025	5.06	2.64	39% - Below average

API/database status: connected

Source: conference paper, Fig. 7 (Low Carbon London run)

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Limitations and future work

- Participant interpretation and SUS usability evaluation remain future work.
- Profile-matched peer groups are not yet implemented.
- Live controller integration and concurrent multi-site ingestion remain untested.
- A configurable per-device power factor and richer tariff models are future refinements.

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Conclusion and future research

- ✓A validated, modular dashboard translates circuit-style data into useful metrics.
- ✓Controlled simulation provides known ground truth; real-data testing supports portability.
- ✓Next: participant usability study, fair peer groups and richer tariff models.
- ✓Further validation should use South African household data and approved live-device feeds.

Thank you • Questions?